

ULAANBAATAR, Mongolia

WMO: 442920

Lat: 47.9183N

Lon: 106.8478E

Elev: 1304

StdP: 86.61

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.2	-34.0	-40.2	0.1	-35.9	-37.9	0.1	-34.0	8.3	-17.9	7.1	-17.7	0.9	170	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	31.0	15.5	28.9	14.8	26.9	14.2	17.7	25.7	16.7	24.5	15.9	23.5	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.1	12.6	19.9	14.1	11.8	19.3	13.1	11.0	18.8	55.3	26.0	52.2	24.2	49.4	23.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.8	7.5	-38.3	34.3	3.9	2.5	-41.1	36.1	-43.3	37.5	-45.5	38.9	-48.4	40.8
			-38.3	19.6	3.9	1.3	-41.1	20.6	-43.4	21.4	-45.5	22.1	-48.4	23.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-23.3	-18.4	-7.9	2.3	9.5	16.1	18.9	16.3	9.4	0.1	-12.2	-20.9	(6)
(7)		DBStd	15.57	4.68	5.98	6.71	5.26	5.06	4.31	3.57	3.68	4.62	4.97	6.86	5.12	(7)
(8)		HDD10.0	4692	1033	794	556	236	72	4	0	2	64	308	666	958	(8)
(9)		HDD18.3	7066	1291	1027	814	482	277	91	34	84	268	567	916	1216	(9)
(10)		CDD10.0	769	0	0	0	4	55	188	277	197	48	1	0	0	(10)
(11)		CDD18.3	102	0	0	0	0	3	25	52	21	1	0	0	0	(11)
(12)		CDH23.3	1519	0	0	0	2	99	409	634	330	44	0	0	0	(12)
(13)		CDH26.7	488	0	0	0	0	22	130	231	100	6	0	0	0	(13)
(14)	Wind	WSAvg	2.6	1.3	1.8	2.6	3.6	3.9	3.5	3.0	2.8	2.9	2.4	1.8	1.3	(14)
(15)	Precipitation	PrecAvg	254	2	2	4	7	22	41	72	60	26	9	6	4	(15)
(16)		PrecMax	377	8	6	20	18	64	84	160	143	70	34	16	8	(16)
(17)		PrecMin	178	0	0	0	0	2	6	8	13	0	2	0	1	(17)
(18)		PrecStd	47	2	2	4	5	18	19	30	27	18	7	5	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.9	0.9	13.0	22.1	29.3	33.2	34.9	32.8	27.1	18.7	7.2	-5.7	(19)
(20)		MCWB	-8.6	-3.4	3.7	8.1	12.1	15.2	17.7	16.4	12.3	7.6	1.1	-7.5	(20)	
(21)		2%	DB	-10.3	-2.8	8.9	18.4	26.1	30.0	31.8	29.2	24.2	15.1	3.9	-8.4	(21)
(22)		MCWB	-11.7	-5.5	1.6	7.1	11.1	14.2	16.5	15.0	12.0	6.0	-0.7	-9.8	(22)	
(23)		5%	DB	-12.9	-5.4	5.9	15.8	23.3	27.8	29.1	27.0	21.9	12.2	0.9	-10.6	(23)
(24)		MCWB	-13.9	-7.5	0.0	5.6	10.1	13.5	15.9	14.7	11.0	4.5	-2.5	-11.6	(24)	
(25)		10%	DB	-15.2	-8.1	2.8	13.1	20.8	25.5	26.9	24.8	19.4	9.7	-1.8	-12.5	(25)
(26)		MCWB	-16.0	-9.7	-1.6	4.3	9.1	13.2	15.5	14.5	10.2	3.3	-4.3	-13.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.7	-3.0	4.2	9.3	13.5	17.2	19.5	18.4	15.0	8.0	1.5	-7.4	(27)
(28)		MCDWB	-7.2	0.3	12.1	20.5	25.7	25.4	28.0	26.5	21.6	17.1	6.9	-5.9	(28)	
(29)		2%	WB	-11.8	-5.4	1.9	7.4	11.9	16.0	18.2	17.1	13.4	6.3	-0.6	-9.8	(29)
(30)		MCDWB	-10.5	-2.7	8.0	17.6	24.0	24.5	26.6	24.5	21.1	14.4	3.4	-8.6	(30)	
(31)		5%	WB	-13.8	-7.5	0.1	6.0	10.6	15.1	17.3	16.1	12.1	4.8	-2.4	-11.6	(31)
(32)		MCDWB	-12.9	-5.6	5.4	14.8	21.7	23.6	25.4	23.4	20.0	11.8	0.7	-10.6	(32)	
(33)		10%	WB	-16.0	-9.7	-1.6	4.5	9.4	14.3	16.5	15.2	10.8	3.4	-4.2	-13.4	(33)
(34)		MCDWB	-15.2	-8.1	2.7	12.5	19.7	22.7	24.3	22.1	18.2	9.1	-1.9	-12.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.6	15.8	15.3	15.5	15.6	13.7	12.9	13.4	15.5	15.1	13.5	12.7	(35)	
(36)		5% DB	MCDBR	15.1	18.0	18.1	20.6	20.9	18.1	17.0	18.4	19.9	19.9	15.7	13.3	(36)
(37)		MCWBR	13.9	15.4	11.7	11.5	10.1	7.1	6.3	7.4	9.7	12.0	11.6	12.2	(37)	
(38)		5% WB	MCDBR	15.0	17.7	16.9	19.0	18.3	14.3	13.4	14.2	16.8	18.7	15.1	13.2	(38)
(39)		MCWBR	13.8	15.2	11.1	10.8	9.3	6.4	5.8	6.5	8.5	11.3	11.3	12.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.264	0.304	0.351	0.382	0.392	0.377	0.357	0.313	0.284	0.258	0.243	(40)	
(41)		taud	2.336	2.291	2.268	2.237	2.190	2.265	2.358	2.389	2.456	2.472	2.396	2.329	(41)	
(42)		Ebn at Noon	857	905	914	898	879	871	879	884	899	876	826	803	(42)	
(43)		Edh at Noon	69	91	111	128	140	131	117	109	92	76	65	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.93	3.09	4.50	5.66	6.50	6.48	6.11	5.42	4.55	3.15	1.99	1.52	(44)	
(45)		RadStd	0.10	0.18	0.19	0.24	0.46	0.36	0.34	0.36	0.25	0.16	0.11	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	-2.30	-3.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page